Question	Answer	Marks	Guidance
3(a)	$H_0: \mu_Y = \mu_X \quad H_1: \mu_Y > \mu_X$	**B1**	If in words, must have **population** means. Allow H_0: $\mu_d = 0$, H_1: $\mu_d > 0$ if d defined or consistent with working.
	Differences: 3, –1, 6, 2, 3, 3, –1, –2, 3, 3	**M1**	Attempt at signed differences, SOI.
	$s_d^2 = \frac{1}{9}\left('91' - \frac{'19'^2}{10}\right) \quad [= 6.1]$	**M1**	Correct form, unbiased, implied by $s_d^2 = 6.1$ or $s_d = 2.47$ seen.
	$\dfrac{\frac{'19'}{10}}{\sqrt{\frac{'6.1'}{10}}} = 2.43$	**M1**	Correct form using *their* s_d^2. Allow M1 for $\dfrac{\frac{'696'}{10} - \frac{'677'}{10}}{'s_p'\sqrt{\frac{1}{10}+\frac{1}{10}}}$ $[= 0.666]$ from 2-sample t-test.
		A1	AWRT 2.43.
	$'2.43' < 2.821$, accept H_0 / do not reject H_0 / not significant.	**M1**	Compare *their* 2.43 with 2.821 and appropriate result (may be in terms of H_1). Allow M1 for comparing *their* 0.666 with 2.552 and appropriate result from 2-sample t-test. Test result may be implied by an attempt at an appropriate conclusion in context ignoring hypotheses. *Their* 2.43 must come from an attempt at standardising.
	Insufficient evidence to support the manufacturer's claim. OR Insufficient evidence to support that (on average) machine Y extracts more juice than machine X.	**A1**	Correct conclusion from correct working ignoring hypotheses. In context. Level of uncertainty in language is used (for example, not 'prove'). Allow 'Not enough evidence / no sufficient evidence that / to show / conclude that…'. Do not accept 'No evidence…'. Do not accept statements such as 'there is sufficient evidence to suggest…'.
		7	

Question	Answer	Marks	Guidance
3(b)	Population of differences is normally distributed. OR Underlying distribution of differences is normal.	**B1**	Population **mean** normally distributed earns B0. (Underlying) **data** is normal earns B0. Must mention **population/distribution** and **differences**. Following 2-sample test in part **3(a)**, allow either of the following: Population**s**/underlying distribution**s** are normal. Populations have equal variances. Must refer to both populations.
		1	
3(c)	The difference remains the same for pair 4. The set of differences has not changed. OR Redo the test: $2.43 < 2.821$	***B1**	Comment that recognises that the differences remain unchanged. OR Compare *their* 2.43 from part **3(a)** with 2.821. Following 2-sample test in part **3(a)**, compare *their* 0.653 with 2.552.
	The conclusion remains the same.	**DB1**	
		2	